Placement Cell 03/10/2011

NVIDIA PREPARATORY QUESTIONS

1. Question on clocks ---see R.S.Aggarwal (3m). Between 4:00 pm and 5:00 pm at what time will the hours and minutes hand be 10 degrees apart for the first time.....????
ans: 4:20 p.m.

2. a)Data compression is used for information storage and transmission. You are to use encryption along with data compression. What will u do?
i)Data compression first and then encrypt the compressed data
OR
ii)encrypt the data and then compress the encrypted data
Justify ur answer.
ans :
Compress the data and then encrypt the data.
Reason : 1) Encryption time is proportional to data size. Compressed data size will be less than normal data size.
2) It also hinders cryptanalysis because there's less chance of repeating patterns and moreover, cryptanalysis is always harder when there is less ciphertext to chew on.

b)Give Binary representation of -125.375 in fixed point 2C notation with 3 bits for fractional part and 8 bits for integer part. (4m)

3. Given a 4 bit binary number design a circuit that gives square of the number.

4. Given a 7 bit binary number design a circuit to find the number of 1s (Binary digit 1s) in the number using only Half Adders and Full Adders.

5. Write a C functon IsLittleEndian() to return true if the machine stores LSB of a Multibyte number in Highest Address and false if it is the other way (Iam not sure whether it is LSB in Highet addr or MSB in highest Address ,check what is LittleEndian and BigEndian).

6. A question on FSM (lengthy Question 8 lines) I don't remember it .

7. A question on FF timing
Two FFs with combo logic blocks in B/w FF1 and FF2 with different delays and combo block b/w o/p of 2nd FF and i/p of 1st FF (feedback). Setup time ,hold time clock-to-Q delay were given

a)with a SKEW of "DELTA" what will be the maximum clock frequency (numerical values were given).

b)for a hold time of "HOLD" ns what will be the maximum SKEW allowed????

8. A question on FSM design (it's a ARBITER problem Problem statement was given along with waveforms describing the function) You are required to draw the FSM showing expressions for state changes and output.

9. a) A binary number is enormously large, so it is divided by a fixed number and the remainder is used.What is this called and why it is called so????
9. b)Give two advantages and one disadvantege of Latch based designs over register based designs.

10. This is a Q from Low power design (circuits). The technology used for a design is changed from 0.13 Um to 0.09 Um and VDD from 1.2v to 1.0 v. If Area of a chip is 81mm2 and Power consumed is 1w and maximum frequency is 500 MHz,in the earlier 0.13 Um technology, what will be the Power consumed in the new Technology of 0.09Um and Vdd of 1.0v? what will be the area of the chip? What will be the maximum frequency if the Power consumed is kept same ???

11. Room lighting.. Room with 100 bulbs and switches.. 100 people goes in one by one. 1st person toggles all switches, 2nd person toggles all switeches which r multiles of 2 (2,4,6..) And so on (till 100th person toggling only 100th switch) once finished with all , which all bulbs r on..

12. Ieee single precision format for 1/5? Form : 1 signbit, 8 bits for exponent, 23 bits for mantissa
13. Elevator of 60 steps, one kid going up also climbs 15 steps during the upward motion. Another kid runs up in the downward elevator and reaches top along with the first kid ?? How many steps are covered by second kid?

(answer = 45 + 60 = 105)

14. Find the minimal exp for a > b, where a=a0a1 and b=b0b1.
15. Find an fsm for any no which is divide by 3.
16. One c ++ class program.
17. One digital logic ckt problem. In signal , clk and out signal are given.. Give the ckt..
18. One cpu-cache hitting problem..
19. Shift register.. (b0b1 selection bits..
    00 - shift lft
    01 - shift right
    10 -
    11 - toggles q1 q2 (hint : uinversal shift register)

20. a) To construct a 2^n :1 MUX. how many number of 2:1 MUXes are required.
b) If propagation delay of each MUX is 1nsec what will be the propogation delay of the above built ciruit.
c) Can your ciruit operate at 1MHz? If not what are the changes to be made?

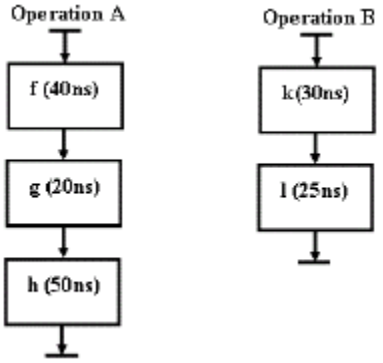
21. Some question on Address banks and had some figure.

22. How to identify a singly linked list that whether it is circular or not?
23. Another question on digital electronics made of delay FF and asyncrhonous input.

24. It consisted on 1register followed by logic cirucit and another register and another logic circuit. Which is feedback to first register.
There were given some Setup time and hold time for register (i can't remember the values) and the logic ciruit were given with max and min delay
Find out the mins. freq of operation.

25. Aptitude question: 2 Races held for A, B, C and D Some conditions given and find out the positions.

26. You are designing a circuit that implements two operations A and B as shown below.

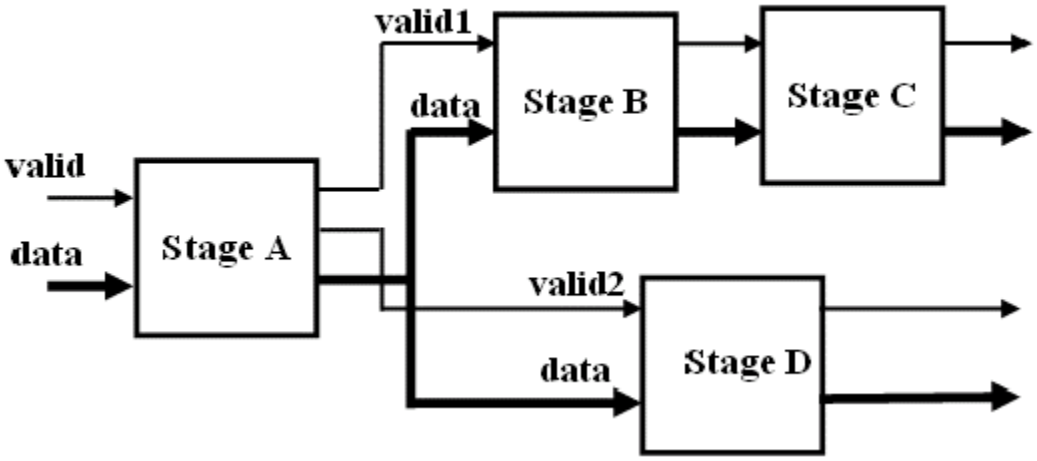


NOTES:

1. At any point in time the circuit is doing either A or B.
2. Delays through modules of each operation are given in the figure below.
3. The circuit must have registers on all inputs. No registers are needed on the outputs.
4. The delay through a register is 5ns.
5. Operation A occurs 70% of the time while operation B occurs 30% of the time.

What is the clock period that will result in highest overall performance?

27. Two stages are added to the pipeline discussed in the earlier post to form the circuit shown below.



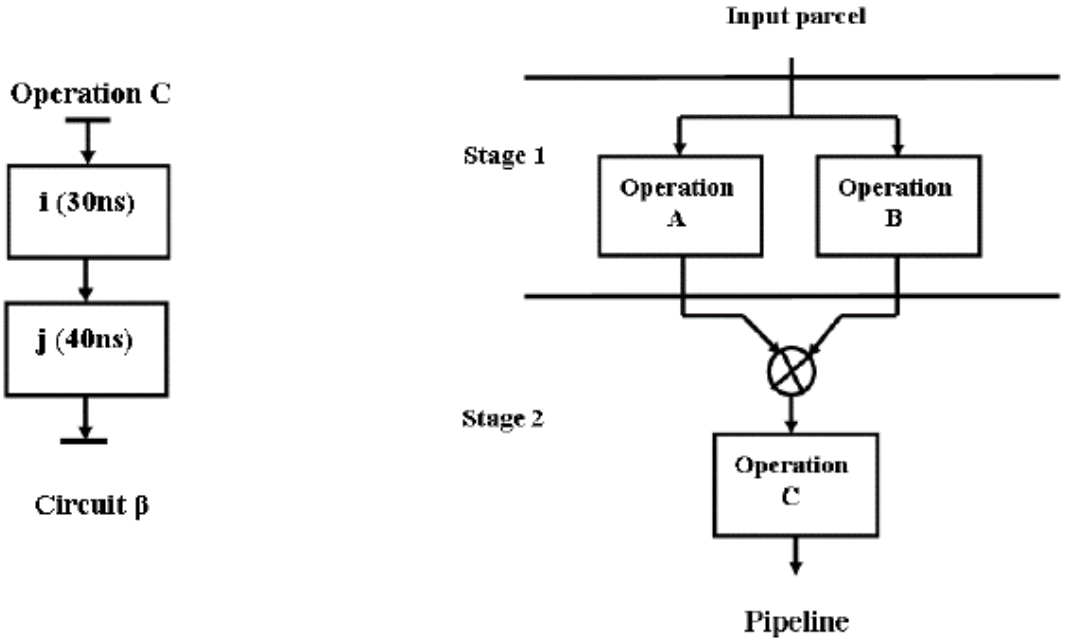
	Stage A	Stage B	Stage C	Stage D
Activity Factor	0.4	0.3	0.45	0.5
Capacitance	10	40	60	15

2. Stage A decides to send the input either to stages B-C or to stage D.
3. 35% of data from stage A is sent to stages B-C while 65% of data from stage A is sent to stage D.
4. All information given in the earlier post is applicable to this pipeline as well.

Extend the power reduction scheme that your developed in the earlier post and apply it to the new pipeline. Calculate how much power will be saved when applying this scheme to the new circuit.

http://digitalelectronics.blogspot.com/2008/01/nvidia-interview-question_6008.html

28. Circuit below, which implements operation C, has been designed such that it has two modules i and j, with the delays shown in the given figure below. The latency for circuit b is 2 and the delay through a register is 5ns. Assume that circuits a and b are stages of a pipeline as shown in the figure below.



NOTES:
1. The pipeline has 2 stages where the first stage of the pipeline contains circuit a (executing operation A or B) and the second stage of the pipeline contains circuit b (executing operation C).
2. Stage 2 reads the output of stage 1 as soon as it is available even if the two operations of stage 1 (i.e. A and B) have imbalanced latencies.
3. The pipeline has registers only on the inputs of the stages
4. The performance and cost overhead due to control circuitry (e.g. Mux) is not considered in this preliminary analysis.

Based on your analysis in the earlier post, what is the total time required to execute 100 input parcels sent through the pipeline?

NVIDIA Interview Questions

1. RTL simulation is faster than delta-cycle simulation but can not be used in all situations.
2. When doing RTL design, all flip-flops need to have a reset input that is synchronized with the clock.

3. If P is a 16-bit unsigned signal, testing if P = 656 requires a minimum of 16 FPGA cells each having an LUT with 4 inputs, 1 output, a carry-in, and a carry-out.
4. Because there are two types of memory operations (Read and Write), there are four different types of data dependencies that can exist between memory operations.
5. When doing RTL design, a HLM can be written only after input/output allocation is done.
6. If the hold time of a flip-flop is violated, a possible solution would be to add buffers at the input of that flop.
7. Voltage scaling is a power reduction technique that relies on reducing the supply voltage of a circuit without affecting any of the other circuit parameters.
8. Because of the small number of transitions between codes, a 32-state finite state machine that uses Gray coding will consume less power than one that uses binary coding.
9. If a circuit contains some redundant components, all faults in the redundant circuitry are undetectable.

http://digitalelectronics.blogspot.com/2008/01/nvidia-interview-questions.html